

YADHU KRISHNAN VB

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OBJECTIVE

- Chemistry Graduate(Bachelor of Science) with strong interest in Chemistry, Materials Science and Biology. Seeking a challenging internship opportunity to apply my knowledge and skills and to gain practical experience in a multidisciplinary environment.

WORK EXPERIENCE:

Research Intern

Alstonia-Simisco Biologics c/o SciGenom Labs Pvt Ltd, Kochi

August 21, 2024 – May 30, 2025 (9 months)

- **Project:**"Assay Development to Measure the Hydrophobicity of Antibodies Using a Fluorescent Dye"
- Developed and standardized a fluorescence-based assay as an alternative to traditional Hydrophobic Interaction Chromatography (HIC) for characterizing monoclonal antibody (mAb) hydrophobicity.
- Gained hands-on experience in therapeutic mAb production, purification, and characterization.
- Carried out experimental planning, execution, and data analysis using software including GraphPad Prism.
- Interpreted results relevant to protein chemistry and the biopharmaceutical industry.
- Contributed to a novel research area with potential for publication (manuscript under preparation).

EDUCATION:

MSc in Data Science and Bio AI

Digital University Kerala, Thiruvananthapuram, Kerala, India

2025 – Present

Bachelor of Science in chemistry

Mahatma Gandhi University/Newman College, Thodupuzha, Kerala, India

June 2021-may 2024

- **Relevant coursework:** Fundamentals of Chemistry, Organic Chemistry, Physical Chemistry, Inorganic Chemistry, Environmental Chemistry, Nanochemistry and Nanotechnology

- **Project coursework:** Green Synthesis and Characterization of silver nanoparticles using Clitoria Ternatea.
 - ✓ Our study was aimed to develop naturally derived, non-toxic, plant-based nanomaterials for antimicrobial activity. The material was synthesized using the extract of Clitoria Ternatea flower, known for its health benefits
 - ✓ Using NaOH, AgNO₃, and the powdered flower, various experiments were conducted to understand the formation of nanomaterials.
 - ✓ The formations of nanomaterials were characterized via the development of brown color formation, and UV-visible spectroscopy (440nm).
 - ✓ This work provides insights into the development of potential antimicrobial nanomaterials using green synthesis methods (plant extracts) and non-toxic materials. Future steps include further characterization and testing the functionality of the materials for their application

Industrial visit:

Project Industrial visit: Kerala Minerals and Metals Limited (KMML)

- Location: Sankaramangala, Kollam, Kerala
- Objective: Gain practical insights into industrial processes and operations

Overview:

- **Company Profile:** KMML is a world-class producer of mineral and sand-based value-added products. It is the only producer and distributor of rutile grade Titanium Dioxide (TiO₂) by chloride process in India.
- **Learning Highlights:**
 - **TiO₂ production:** Observed the chloride technology process for manufacturing Titanium Dioxide pigment.
 - **Process Steps:**
 - ◆ Reduction and leaching of raw limenite containing 55-60% TiO₂.

- ◆ Regeneration of spent hydrochloric acid to enhance economic efficiency and prevent pollution.
- ◆ Conversion of beneficiated limenite into TiO₂ pigment.

Skills Acquired:

- Understanding of industrial-scale chemical processes.
- Knowledge of environmental and economic considerations in industrial production.
- Exposure to real-world applications of academic concepts.

Skills:

Software skills

- MS-Word
- MS-Excel
- MS- Power point
- MS- OFFICE

Soft skills:

- Leadership: Developed through leading NCC activities and team
- Teamwork: Collaborated effectively with peers during NCC camps and projects
- Discipline: Enhanced through structured training and participation in NCC programs
- Time management: Balanced academic commitments with NCC responsibilities
- Communication: Strengthened through group activities in NCC

LANGUAGE :

- Malayalam
- English
- Hindi